nlcsplit: fragment and pair resolution of the VV10 nonlocal-correlation energy

Angran Xia*
School of Chemistry & Chemical Engineering, Shaanxi Normal University, Xi'an 710119, Shaanxi, China
haangyeon@snnu.edu.cn

Changwei Wang
School of Chemistry & Chemical Engineering, Shaanxi Normal University, Xi'an 710119, Shaanxi, China

Draft — 29 September 2026
ORCIDs PENDING, author-supplied

*SoftwareX / JOSS shape. Every numerical claim carries a `[S#]` tag into this paper's own `## Appendix: sources`, which resolves each tag to a tracked artefact in this repository — not to `paper/main.md`, whose tags number a different set. Archived stdout lives under `evidence/**` with SHA-256 manifests; `nlcsplit/logs/` is gitignored scratch and is not a source. **The front matter contains fields only the author can supply (ORCID, corresponding status, co-author confirmation); they are marked rather than filled.** `references.bib` is the shared list; the entries this paper cites were reverse-checked against Crossref by DOI, not by title search (see `RESEARCH-20260929-lit.md` §9 for why that distinction mattered, and `evidence/bib_provenance/G_s22_provenance_20260930.md` for the queries behind the two software entries).*

1 Abstract

VV10-type nonlocal correlation is evaluated in production electronic-structure codes by collapsing a double sum over grid-point pairs into per-point values, which discards the pair index. `nlcsplit` evaluates that double sum explicitly and attributes each grid point to atoms and fragments through a real-space partition, so the nonlocal contribution separates into intra-fragment and inter-fragment pair blocks that sum back to the total to machine precision. The package is pure Python and NumPy [1] on top of PySCF [2]; [3], released under Apache-2.0, contains no GAMESS source, and reproduces every number it documents from a single named script.

2 1. Motivation and statement of need

Functionals that pair a range-separated hybrid with the VV10 nonlocal correlation term - ωB97X-V [4] and ωB97M-V [5] in particular - are a routine choice for non-covalent interactions. Their nonlocal contribution,

$$E_NL = \frac{1}{2} \int\int \rho(r)\ \rho(r')\ \Phi(r, r')\ d^3r\ d^3r',$$

*Corresponding author; e-mail in the byline.

is a two-centre sum, but what users get back is one number per grid point, or one total. When an interaction energy needs to be interpreted per fragment, the available options stop at the semi-local part: electrostatics, exchange, polarisation and charge transfer can be resolved by EDA schemes, whereas the nonlocal term is carried as an undivided block. The closest published treatment computes exactly that block as a whole and asks whether it is accurate [6]; the closest published explicit evaluation, JuNoLo [7], sums the double sum directly but targets an improved total energy and explicitly recommends itself for bonded systems "where one cannot tell apart fragments". Neither resolves the term into fragment pairs, which is what the literature screen of [S1] was run to find - and that screen's own record says the open-index *subject* search behind it is unreliable, so this is a screen, not an enumeration. That gap is what this package fills, and it is a capability gap rather than an accuracy gap: on the same grid and density our sum agrees with the native PySCF kernel to ~1e-8 Ha [S9].

3 2. Method overview

Kernel and convention. `w0 = sqrt(C|∇ρ|^4/ρ^4 + 4πρ/3)`, `κ = b(3π/2)(ρ/9π)^(1/6)`, `g = w0(r_i)R^2 + κ(r_i)` with same-centre pairing, `Φ = -3/(2 g g' (g + g'))`, and `E_NLC = ½∫∫ρρΦ + β∫ρ` with `β = (1/32)(3/b^2)^(3/4)`. Parameters are read from PySCF `nlc_coeff` rather than hard-coded. Same-centre pairing is the convention that reproduces the native kernel; the cross-pairing variant `g = w0(r_j)R^2 + κ(r_i)` is retained as a definition-sensitivity control and raises monomer energies by 0.138 (water), 0.418 (argon) and 0.914 (benzene) kcal/mol [S5].

Blocks and additivity. `pair_energy(A, B)` returns the half block ½ Σ_i∈A Σ_j∈B; the inter-fragment term is reported as twice that, so `E_nl = Σ_A intra[A] + Σ_A<B inter[A,B]` holds as an identity. Measured residual: 0.0 to 3.5e-18 Ha [S2].

Why decomposing only the double sum is exact. β is a constant and $\int\rho$ is the electron number, so the local term $\beta\int\rho = \beta$N cancels identically from any superposition difference. Inter-fragment decomposition therefore needs no convention for splitting a local energy density: it is a property of the double sum alone [S3]. The flip side is reported as a limitation rather than hidden - the absolute local term of one fragment is exactly $\beta\cdot$N_A, so its scheme spread is β times the spread of the Becke population, 3.0415 kcal/mol per electron, i.e. 0.329 kcal/mol for the water dimer on the official S22 geometry and 0.300 on the constructed one (`nlcsplit/step_k_population.py`, whose two runs both assert that the fragment populations sum to the electron number).

Attribution. Four real-space partitions reachable through the PySCF grid machinery - Becke, Becke with the atomic-radius adjustment, Becke-Treutler and Stratmann - plus a nearest-atom hard assignment on a common grid. A point may be shared by more than one fragment; the factor matrix is exposed as `partition_factors(mol, level, scheme) -> (coords, base_w, S[N, natm])`, normalised so that `Σ_a S[i,a] = 1` pointwise. The function refuses to return unless two internal assertions pass: normalisation, and `base_w · S[i, owner(i)]` reproducing the weights that PySCF's own `get_partition` produces - an external anchor against the same C kernel PySCF uses, not against a re-implementation in this package. The two checks it enforces are pointwise normalisation `|Σ_a S[i,a] - 1| < 1e-12` and agreement with PySCF's own weights `max|base_w·S[i,owner(i)] - w_ref| < 1e-11`, with a normalised mean deviation below 1e-14; the tolerances are the assertion thresholds in the published test, not residuals we are claiming to have measured [S17].

Far-field cut. Points below `rho_floor = 1e-12` are dropped from the double sum. Sweeping 1e-8 … 1e-16 changes the inter-fragment term by ≤5e-11 kcal/mol while discarding 10.7-22.6%

of points carrying ≤2.3e-6 electrons [S7], so the threshold is an implementation detail rather than a parameter.

4 3. Software description

nlcsplit/ nlc.py VV10 fields, pair_energy (gemm path), fragment_decomposition partition.py four attribution schemes, partition_matrix, partition_factors geomlib.py single source of geometry truth; internal-coordinate assertions at construction; covalent-connectivity fragment splitting; official GMTKN55/S22 loader step*.py one script per table in the paper; stdout archived UTC-stamped figures.py figures built from the JSON sidecar, never from log text tests/ kernel, regression, geometry and partition-factor suites examples/ quickstart plus two comparison checks

Dependencies: NumPy and PySCF. `partition.py` reaches into `libdft.VXCgen_grid` through ctypes deliberately - it is the entry point that `gen_grid.get_partition` itself uses, which is what makes the audit above external rather than circular.

Representative output, produced by `nlcsplit/examples/quickstart.py` (ωB97X-V/6-31G*, water dimer, grid level 1, kcal/mol):

inter[(0,1)] = -0.5237 intra = -3.5457 -3.7437 E_nl = -7.8130 E_loc = 60.8303 E_total = 53.0172

`E_nl` equals intra + intra + inter as required, and `E_total` may be compared with the native kernel evaluated on its own reference grid, which gives 53.017 and 53.016 for this system and level (difference +0.002 kcal/mol, which is the grid representation difference of section 3.1 of the companion text rather than an algorithmic one) [S9].

5 4. Impact

The package makes three things checkable that previously were not. It gives a number for how much of the nonlocal correlation is inter-fragment, which on the six S22 complexes that carry a counterpoise reference at grid level 2 accounts for 0.92-1.08 of the frozen-density superposition difference - association-time nonlocal correlation is almost entirely carried by point pairs straddling the fragments [S2, S10]. It supplies the per-system uncertainty that the attribution choice contributes (0.004-0.247 kcal/mol absolute, 0.8-18.2% relative, with no ordering by interaction type or by system size), which is the error bar any future use of such numbers must carry [S12]. That span is not basis-set invariant: on the water dimer it is 0.0249 kcal/mol in STO-3G against 0.0382-0.0400 in the four larger sets, so the number moves by ~38% between STO-3G and 6-31G* while the four sets beyond STO-3G agree with each other within 5% [S12]. Nor is it grid-converged to better than it is printed: level-2 values are stored to three decimals, so a level-1 to level-2 change below 5e-4 kcal/mol is not resolvable in the archived table, and the one four-decimal ladder we ran gives 0.0002 kcal/mol between levels 1 and 2 - small, but not zero. And it produces evidence about the quantity itself: where the share is defined at all it ranges from 16% of ΔE_xc (water) to 97% and 52% for the two geometries of the benzene dimer, and on the methane dimer the two parts carry opposite signs so the package's own rule ($|\Delta E_NLC/\Delta E_xc| > 2 \Rightarrow$ report the two absolute terms, not a ratio) refuses to state a share at all. The VV10 term therefore cannot be reported as "the dispersion energy" without qualification, and on one row of our own table not even as a percentage [S2].

6 5. Availability and quality assurance

Apache-2.0; no GAMESS source is included, linked or required. That claim is checked by scanning the published tree for GAMESS *source artefacts* specifically - `find` over `*.src`, `*.f`, `*.for` and `GAMESS*` returns 0 - and not for the word. An earlier version of this sentence asserted that the word appears only in the package docstring, and that assertion was false: 13 files contained it, 8 of them as part of a working-machine path (`/root/gamess_work/...`, `/mnt/e/Project/GAMESS/...`) including inside the shipped `step_h.json` and `run_all.sh`, which is how a repository name that reads as "this is the GAMESS project" got into a release that claims not to contain it. Two defects were behind that: the gate's patterns covered artefact extensions but not path forms, and it iterated the list of copied names rather than the directory tree, so 13 `.pyc` files that pytest had left inside the release directory were never opened at all. Both are now fixed - the gate walks the tree and fails by default on path forms, bare cluster addresses and account names - and the published copies are scrubbed of those paths while the numbers are left untouched, with a byte-level ledger of every replacement kept beside the release and the archived stdout in this repository still hashing to what was recorded for it. After the fix the word appears in 5 published files, all of them the disclaimer or prose about it, and the path forms in 0. Two further defects were found while re-checking this section on 2026-09-29, and both were of the kind a source-tree reading cannot see. The first: an earlier build was reported fixed and still shipped two script-style `test_*.py` modules at package top level; the wheel now has 36 members and none of them is a script-style test - it grew again when `step_l_sweep_compare.py` was added to the whitelist, see below. The second is worse and was found only by looking at the artefact: **`fastpair.py` was not in the package at all**, while the shipped `step_j_scaling.py` imports it - so a reader installing the wheel could not run a single number from the companion algorithm paper, and gate E (which checks that every import is *declared*) stayed silent because the import is intra-package. `fastpair.py` and its test module are now in the manifest. In the same pass `test_tolerance_probe.py` was published too, because a tolerance tightened from 0.10 to 1e-4 kcal/mol is a claim about measured spread and the measurement belongs beside it.

A third omission of the same kind was found by asking the question the other way round: `run_all.sh` - the advertised one-command reproduction path - knew four of the thirteen `step_*.py` scripts, and `step_l_sweep_compare.py` was missing from the shipped step list altogether. That script is the only producer of the "agrees to 2e-3 kcal/mol with an independently written script" sentence in section 2.1, so until now the published tree contained neither that script nor any artefact carrying the comparison column. The driver now lists all thirteen with each one's precondition (which need an argument, which need a geometry download, which need an unloaded machine), derives the repository root from its own location instead of a hard-coded working path, and refuses to run the argument-taking ones rather than silently running them at their defaults.

The suite collects 64 tests in the working repository - 44 under `-m "not slow"` and 20 under `-m slow`, a partition of the same collection and not two numbers to be added; the whole collection is on record green at 811 s. The published wheel ships six test modules plus `__init__.py` and `conftest.py`, which collect 44, and the 20 that stay behind are exactly `test_fastpair_ide.py`'s - the peer's frozen control module, whose `fastpair_ide.py` is deliberately not distributed, since shipping the test without the module would be a collection-time ImportError. Two different out-of-tree runs must not be quoted as one: `pip install .` from the source directory in a clean environment passes 62 with one skipped (the S22 geometry fetch, by design), because that install pulls in the whole tree including the control module, whereas `pytest -pyargs nlcsplit` against the built wheel, run from a directory that is not the source tree, exercises only the published tests - 42 passed and one skipped, or 24 passed and one skipped under `-m "not slow"`. Those two

archived figures are quoted at the collection size they were taken at (43 published tests), because the 44th - a test of the grid measure behind `partition.partition_factors`, added 2026-10-01 - postdates both logs; the run against the wheel rebuilt by that same export is archived beside them with its own exit codes rather than folded into the older sentence. The full out-of-tree run is on record twice, at 581 s and at 407 s, because the wheel is rebuilt by every export and its SHA-256 therefore changes even when no packaged byte does; both logs, both hashes and the shell-quoting bug that left the hash field blank in one of the two are in `evidence/release_qa/` [S19]. Tables map to one `step_*.py` each; the archived stdout a claim cites lives under `evidence/**` with a UTC-stamped filename and a hash manifest, while working logs under `nlcsplit/logs/` are gitignored scratch and are not cited as evidence. Including a naive triple-loop versus BLAS comparison of the kernel, per-scheme additivity checks, and geometry guards that assert bond lengths and angles at construction time.

Thirteen export gates run locally on every rebuild (`node tools/make_public_repo.js`, which ships in the published tree). They are not all of equal standing, and the difference matters: some exist because they caught a real defect, others only because we could not otherwise show the earlier ones were not vacuous. The last column says which, and "planted" means the gate has never fired on real input - it fires on a fault we put in deliberately to prove it would.

Gate D exists because the first three could only fail in one direction. A, B, C catch material that should not be present; what actually went wrong repeatedly was material that should have been present and was not - a cited stdout living only on a remote machine, calibration scripts inside a gitignored directory. Those produce a clean export and a reader holding a citation to nothing. Gate E has the same blind spot in the other direction and we hit it: it checks that an import is *declared*, so it says nothing when a shipped module imports a sibling that was never packaged - the `fastpair.py` defect above.

The gate code itself now ships, and that exposed a self-reference. `tools/make_public_repo.js` and `tools/audit_no_gamess.js` are published under `tools/` because this section and `CONTRIBUTING.md` tell the reader to run them; until 2026-09-30 they were not, so that instruction was executable only on the authors' machine - the same class of defect as a citation pointing into a gitignored directory. Publishing a scanner, however, publishes what it scans for: the exemption ledger printed by the first build that contained the scanner listed 54 lines, and every one of them was a detection pattern naming a cluster internal address, an SSH port, an account name or a working directory. Those ten rules, together with the fault fixtures that exercise them, now live in `tools/intranet_patterns.private.js`, which is excluded by design and named in the export report. The published copy keeps the shape-based rules (credential assignment, private-key block, `ssh ... @`, bare-IP user), loads the named rules when they are present, and prints `note_private_rules_loaded`; a run that cannot load them is not reported as clean - it says so, and gate B is then checking four rules rather than fourteen. Running the published copy against a copy of the published tree turned up two more holes of the same kind: rebuilding crashed on the two appendix sources that are renamed on the way out, and the proxy scrub was not idempotent, stacking its own replacement comment a second time. Both are fixed, and a second run over the same tree reproduces the first byte-for-byte except for the export timestamp (`evidence/release_qa/`) [S19]. `tools/audit_no_gamess.js` makes its single legal GAMESS hit a named rule in which the file *and* the line must match, so a non-disclaimer sentence inside the same file is still a violation, and it refuses to scan a directory that was not produced by a real install, since reading a stale copy of the build directory is how an earlier version of this check passed.

Installing from the source tree is not the same act as installing the release, and we had been describing it as if it were. `pip install .` collects every module inside the package directories; the export whitelist does not constrain it. An install from the source directory collects 37 packaged

gate	what it checks	standing
R	walks the **published tree** and fails by default on working-machine paths, bare cluster addresses and account names	real defect: 13 `.pyc` files pytest had left inside the release directory, never opened by the old word-only check
A	no GAMESS source artefacts (matched on source patterns, not on the word "gamess", which the disclaimer itself contains)	real defect: a permanent false positive on our own disclaimer, which is why it matches patterns and not the word
B	no credentials or internal network strings	real defect: a scrub that re-injected a proxy address into a comment
C	every `evidence/*/MANIFEST.sha256` reproduces byte-for-byte	real defect: a Windows path making `sha256sum` report six mismatches that were not there
C2	the scrubbing changed no digit in any number, **and** every file in a published evidence directory is registered in that directory's own source manifest	the second half fired on real input the day it was written: a comparison JSON produced by a step script landed in `evidence/agentb_sweep/` with no manifest entry, and the build refused to ship it (`shipped: false`)
D	every path cited by the companion text's provenance appendix is actually shipped	real defect: three citations that resolved to nothing in the published tree
E	every import in a shipped module is declared in `pyproject.toml`	real defect: `figures.py` importing matplotlib, which no dependency listed
F	the wheel and sdist member lists are what we think they are	real defect: two script-style tests that survived a build already reported fixed
G	`LICENSE` in the **source** file carries the real copyright line, not the Apache placeholder	real defect: the legal statement existed only in the generated copy; fires today on a restored placeholder
T	nothing in the export tree is missing from the export manifest	added as a belt-and-braces check; no firing on real input is recorded in this repository, only on the faults we planted
U	every module a shipped file imports from `nlcsplit` (absolute, dotted or package-relative) is itself in the shipped tree; the gate also self-tests against a planted missing sibling, and a build where that self-test does not fire does not ship	real defect: it was written for `fastpair.py`, which gate E could not see because an intra-package import is not a declared dependency
V	every Python module inside a packaged directory is either in the export whitelist or named, with a reason, as an unreleased internal module - this is the set `pip install .` actually puts in `site-packages`, which the whitelist does not constrain; self-tests fire on a planted stray and stay silent on the named ones	real defect: found by the shipped licence scanner, which walked 45 installed files from a source-directory install and found the peer implementation among them
W	no manuscript or README quotes a *share* for a system whose denominator cancels, i.e. where ‘	dE_NLC/dE_xc

modules where the wheel ships 31 (the scanner walked 45 installed files in total), and among them it reported `nlcsplit/fastpair_ide.py` - the peer implementation this repository keeps as a frozen comparison, deliberately not released - sitting in `site-packages`. Gate V now fails a build if any Python module inside a packaged directory is neither whitelisted nor named, with a stated reason, as an unreleased internal module; the six currently named are that implementation, its test and its validation script, plus three measurement scripts whose outputs are archived under `evidence/`. The same run also surfaced the provenance record pip writes into its own `dist-info` directory, which stores the *caller's* install path - here a working-machine directory whose name contains the very word being scanned for. It is a named exemption whose predicate strips the URL value before deciding, so anything other than a path inside that file is still a violation.

Release hygiene is now executed rather than specified: a clean-environment install of the source tree, and an out-of-tree `pytest -pyargs nlcsplit` against the built wheel from a directory outside the repository, both recorded with the wheel hash. Writing the specification down is what produced two of the fixes: script-style `test_*.py` modules were removed from the package (pytest collects and executes them at import), and the `slow` marker was registered in package configuration rather than only in repository configuration. An earlier out-of-tree import by path - weaker, since it read the exported tree rather than an installed distribution - found that `step_i_gridrepr.py` ran a full self-consistent-field sweep at import time because it had no `__main__` guard, since fixed.

The continuous-integration workflow is written but has **never executed**, because there is no remote repository yet. Reviewers should read it as intent. Writing it down did produce one real fix: its PySCF version matrix interpolated as `pip install "pyscf==>=2.13"`, which is not a valid requirement, so the second job would have died at the install step.

Independent verification of the arithmetic now has a second program behind it. An earlier version of this documentation withdrew an ORCA agreement because re-reading the archived output found no separately reported nonlocal term - and that withdrawal was itself wrong: the quantity is printed, under the label `NL Energy, E(C,NL)`. Grepping the same output for "nonlocal" or "VV10" is not a way to find it, but not for the reason we first gave: the words are there, in the run banner's feature list (`NL-DFT gradient (VV10)`, line 116 of `evidence/orca_run/h2o_dimer.out`), which prints on every run and reports no number. The `E(VV10-nl)` column that does match belongs to the PT2 correlation of double-hybrid functionals, which is a different physical quantity. With the label known, twenty-four ORCA 6.0.1 single points covering all eight of the reported systems agree with ΔE_NLC to within 1% of the term everywhere (0.0004 kcal/mol on the water dimer up to 0.0416 kcal/mol on the benzene dimer A), with the outputs and their SHA-256 manifests in the repository [S18]; the comparison is regenerated from those artefacts by `evidence/orca_widen/generator_cross_program_table.py`, which refuses to emit the table if the system list or any row's basis, grid level or counterpoise setting disagrees. A second program can check what the term is worth; it cannot check who owns it, because it does not attribute. That gap is now measured too, in `evidence/attribution_referee/`: on a two-Gaussian density an exact reference cross term is built that contains no partition, no grid assignment and no fragment ownership anywhere in its definition, and evaluated three independent ways - a 3-D Cartesian tensor product, a 5-D cylindrical reduction about the internuclear axis, and importance-sampled Monte-Carlo with its own standard error. Handed the exact density factors, the package's own summation code reproduces that reference to `7.96e-07` Ha worst, inside the reference's combined measured error bar of `1.060e-05` Ha; handed the production Becke/owner assignment, the same code sits `1.7e-08` to `1.26e-04` Ha away from it, and the offset does not move when the node set is changed, so it belongs to the convention rather than to the grid [S21]. The referee states its own resolution rather than hiding it: the offset exceeds the error bar in 5 of its 12 configurations, and in the other 7 it is smaller than the reference's uncertainty, which is reported as "cannot distinguish",

not as "agrees". Two limitations there are unchanged. A third, fully independent implementation (Psi4, or libxc's `xc_nlc_vxc` through its own bindings) is still outstanding - the point is that ORCA is the *only* second program, not that there are three. And the scope of what is checked: on the water dimer the comparison covers the absolute term for the complex and for both monomers, while for the other seven systems the second program checks the superposition difference - which is the quantity every published claim here is made of, but not each absolute per-fragment value.

7 6. Limitations

Quadratic cost in retained grid points, with no sub-quadratic approximation. One attempt at a certified screening was built and measured, and rejected - not on the weak version of the argument we first wrote here. The earlier sentence claimed the bound prunes nothing "at the tolerance the companion text needs (1e-8 Ha absolute)"; that mis-states both sides. `1e-8 Ha` is a per-node-pair acceptance constant, and the companion text needs 1e-6 kcal/mol on a fragment-pair term, which is 1.6e-9 Ha - so the constant and the requirement were never the same kind of quantity. Measured properly, three things hold: at atom-block granularity the *smallest* block-pair bound in a real water dimer is 197.96 Ha against a quantity of -8.3e-4 Ha, so nothing is prunable at that granularity whatever the tolerance (0 of 21 pairs, and 0 of 300 on the 24-atom sandwich); refining below the atom block does fire, and at the accuracy the decomposition needs it still evaluates 84.4-95.4% of the far-field point pairs; and with no other runnable user task at the sampling instant (a contention gate, not a claim that the machine was idle), the screened path is slower than the dense sum in every one of 33 timed screened/exact pairs across two independent windows on the same box - 1.08-10.28x in the archived window and 1.37-13.29x in the second (1.95-10.28x and 2.63-13.29x at the tightest tolerance tested), with no faster sample in either. The radius the bound subtracts is set by the far tail of each atom's density, so the certified separation comes out zero for every pair, including the ones facing each other across the inter-fragment gap. That is a statement about this particular bound, not about screening in general, and the measurements are in `paper/p2.md` with their artefacts under `evidence/bound_forms/`, `evidence/fastpair_ab/` and `evidence/scaling/` rather than quietly dropped [S20]. Interpretation limited to the two-body and fragment-pair level. Results depend on the choice of real-space partition to the extent quantified in section 4. VV10 only - the τ-dependent rVV10 form is not implemented. And it is not an energy decomposition analysis: it does not separate electrostatics, exchange-repulsion, polarisation or charge transfer.

8 Appendix: sources

Paths are relative to the repository root, because that is what the export gate resolves them against. Every path below is tracked by git, and the build refuses to produce `software.tex` if one is not. Three rows used to name `logs/<something>.out`, which is not a path in this repository: `nlcsplit/logs/*.out` is gitignored scratch, and those files are tracked only as the hash-manifested copies under `evidence/**` (the build does place a scrubbed copy of two of them inside the package, but a citation should point at one tracked original, not at a generated duplicate). The two `.json` logs in the S2 row are tracked in place, and are cited as they are kept.

Literature is not in this table. Where a sentence needs a published work it uses a `[[key]]` citation resolved against `paper/references.bib`, whose entries were reverse-checked through Crossref by DOI; the queries for the software and benchmark entries are archived in `evidence/bib_provenance/G_s22_provenance_20260930.md`.

Tag	Artefact	What it fixes
S1	`RESEARCH-20260929-lit.md` §1, §6, §9	the literature screen behind the statement-of-need comparison, together with that file's own measurement that open-index *subject* search recall is poor — which is why §1 calls the gap a screen and not an enumeration
S2	`nlcsplit/logs/step_h.json`, `nlcsplit/logs/step_h_l2cp.json`, `MAIN-skeleton.md` Table 2 (uncertainty decomposition), Table 3 (P_frz against the superposition difference) and Table 4b (the S22 subset)	the block-additivity identity and its measured residual (0.0 to 3.5e-18 Ha), and the per-system tables the decomposition claims to reproduce
S3	`nlcsplit/step_f_dispersion.py`	that $\beta\int\rho$ `=` β`N` cancels identically from a superposition difference, so the inter-fragment term needs no local-density convention — and the denominator rule that refuses to quote a share when the sum cancels
S5	`nlcsplit/step_e_kernel.py`	the definition-sensitivity control: cross-pairing instead of same-centre pairing raises monomer energies by 0.138 (water), 0.418 (argon) and 0.914 (benzene) kcal/mol
S7	`nlcsplit/step_d_floor.py`	the `rho_floor = 1e-12` far-field cut, and the reported cost of it (10.7–22.6% of points dropped, carrying $\leq$ 2.3e-6 electrons)
S9	`nlcsplit/step_i_gridrepr.py`, `evidence/recompute/20260929T_step_i_gridrepr.out`	agreement of the explicit pair sum with PySCF's native VV10 kernel on a fixed grid, and the grid-representation residual quoted in §1 and §2
S10	`evidence/l2cp/20260928T213839Z_centos-step_h_l2cp.out`, `evidence/l2cp/20260929T051912Z_centos-l2_benzmethstep.out`	the level-2 counterpoise numbers, computed on the cluster and archived with their stdout
S12	`evidence/recompute/20260929T003144Z_stepg-recompute.out`	the basis set and grid sensitivity of the attribution, including the ±5% agreement among the four basis sets beyond STO-3G
S17	`nlcsplit/tests/test_partition_factors.py`	the two assertions `partition_factors()` enforces before returning (pointwise normalisation at 1e-12, agreement with PySCF's own `get_partition` weights at 1e-11) — assertion thresholds, not measured residuals
S18	`evidence/orca_run/`, `evidence/orca_control/`, `evidence/orca_widen/`, `evidence/orca_widen/cross_program_table.txt`	the ORCA 6.0.1 outputs and their manifests (`evidence/orca_run/MANIFEST.sha256`, `evidence/orca_control/MANIFEST.sha256`, `evidence/orca_widen/MANIFEST.sha256`): six runs for the water and benzene-A anchors, the ωB97X control group, and eighteen more covering the remaining six systems, with the archived stdout for every one of the 24; plus the two generators that re-derive the comparison from those outputs and refuse if the system list or any row's definition disagrees
S19	`evidence/release_qa/`	clean-environment installs and out-of-tree runs, each recorded with the SHA-256 of the wheel that was installed
S20	`evidence/bound_forms/`, `evidence/fastpair_ab/`, `evidence/scaling/`, `evidence/coretail/`	the certified-screening measurements summarised in §6, reported in full in `paper/p2.md`
S21	`evidence/attribution_referee/run.py`, `evidence/attribution_referee/referee_lib.py`, `evidence/attribution_referee/summarize.py`, `evidence/attribution_referee/README.md`, `evidence/attribution_referee/outputs/summary.md`, `evidence/attribution_referee/outputs/results.json`, `evidence/attribution_referee/outputs/results_run1_20260101.json`, `evidence/attribution_referee/outputs/table_run1_20260101.md`, `evidence/attribution_referee/MANIFEST.sha256`	the partition-free referee for attribution: the two-Gaussian model density, the three independent evaluations of the exact cross term, the measured error bar it is compared against, and the two shipped runs whose tables are byte-identical apart from recorded wall-clock fields. Re-running `evidence/attribution_referee/run.py` regenerates everything; `evidence/attribution_referee/summarize.py` re-quotes it with no re-quadrature

References

[1] Charles R. Harris, K. Jarrod Millman, Stéfan J. van der Walt, Ralf Gommers, Pauli Virtanen, David Cournapeau, Eric Wieser, Julian Taylor, Sebastian Berg, Nathaniel J. Smith, Robert Kern, Matti Picus, Stephan Hoyer, Marten H. van Kerkwijk, Matthew Brett, Allan Haldane, Jaime Fernández del Río, Mark Wiebe, Pearu Peterson, Pierre Gérard-Marchant, Kevin Sheppard, Tyler Reddy, Warren Weckesser, Hameer Abbasi, Christoph Gohlke, and Travis E. Oliphant. Array programming with NumPy. *Nature*, 585:357–362, 2020. doi: 10.1038/s41586-020-2649-2.

[2] Qiming Sun, Timothy C. Berkelbach, Nick S. Blunt, George H. Booth, Sheng Guo, Zhendong Li, Junzi Liu, James D. McClain, Elvira R. Sayfutyarova, Sandeep Sharma, Sebastian Wouters, and Garnet Kin-Lic Chan. PySCF: the Python-based simulations of chemistry framework. *WIREs Computational Molecular Science*, 8:e1340, 2018. doi: 10.1002/wcms.1340.

[3] Qiming Sun, Xing Zhang, Samragni Banerjee, Peng Bao, Marc Barbry, Nick S. Blunt, Nikolay A. Bogdanov, George H. Booth, Jia Chen, Zhi-Hao Cui, Janus J. Eriksen, Yang Gao, Sheng Guo, Jan Hermann, Matthew R. Hermes, Kevin Koh, Peter Koval, Susi Lehtola, Zhendong Li, Junzi Liu, Narbe Mardirossian, James D. McClain, Mario Motta, Bastien Mussard, Hung Q. Pham, Artem Pulkin, Wirawan Purwanto, Paul J. Robinson, Enrico Ronca, Elvira R. Sayfutyarova, Maximilian Scheurer, Henry F. Schurkus, James E. T. Smith, Chong Sun, Shi-Ning Sun, Shiv Upadhyay, Lucas K. Wagner, Xiao Wang, Alec White, James Daniel Whitfield, Mark J. Williamson, Sebastian Wouters, Jun Yang, Jason M. Yu, Tianyu Zhu, Timothy C. Berkelbach, Sandeep Sharma, Alexander Yu. Sokolov, and Garnet Kin-Lic Chan. Recent developments in the PySCF program package. *The Journal of Chemical Physics*, 153:024109, 2020. doi: 10.1063/5.0006074.

[4] Narbe Mardirossian and Martin Head-Gordon. ωb97x-v: A 10-parameter, range-separated hybrid, generalized gradient approximation density functional with nonlocal correlation, designed by a survival-of-the-fittest strategy. *Physical Chemistry Chemical Physics*, 16:9904, 2014. doi: 10.1039/c3cp54374a.

[5] Narbe Mardirossian and Martin Head-Gordon. ω b97m-v: A combinatorially optimized, range-separated hybrid, meta-gga density functional with vv10 nonlocal correlation. *The Journal of Chemical Physics*, 144:214110, 2016. doi: 10.1063/1.4952647.

[6] Saswata Dasgupta, Etienne Palos, Yuanhui Pan, and Francesco Paesani. Balance between physical interpretability and energetic predictability in widely used dispersion-corrected density functionals. *Journal of Chemical Theory and Computation*, 20:49–67, 2023. doi: 10.1021/acs.jctc.3c00903.

[7] Predrag Lazić, Nicolae Atodiresei, Mojtaba Alaei, Vasile Caciuc, Stefan Blügel, and Radovan Brako. Junolo – jülich nonlocal code for parallel post-processing evaluation of vdw-df correlation energy. *Computer Physics Communications*, 181:371–379, 2010. doi: 10.1016/j.cpc.2009.09.016.